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BATTERY CELL AND MANUFACTURING METHOD OF BATTERY CELL CASE

Abstract

The present disclosure relates to a battery cell comprising a case body accommodating an electrode assembly, a case cover covering at least one face of the case body which is open, and a vent notch portion formed by being recessed at at least one of one surface of the case body or one surface of the case cover. The vent notch portion includes a straight portion having a straight pattern, and a curved portion connected to the straight portion and having a curved pattern. A width of the straight portion is different from a width of the curved portion.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority under 35 U.S.C. § 119 (a) to Korean patent application number 10-2024-0030732, filed on Mar. 4, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] Various embodiments of the present disclosure generally relate to a battery cell and a manufacturing method of a battery cell case.

2. Related Art

[0003] Recently, research on battery cells has been actively conducted. Battery cells convert electrical energy into chemical energy and store it and can be reused multiple times by charging and discharging. Due to their economical and eco-friendly nature, battery cells are widely used in a variety of industries.

[0004] On the other hand, as the battery cells are repeatedly charged and discharged, the internal pressure increases and a large amount of heat is generated, causing a swelling phenomenon. This swelling phenomenon can lead to explosion or fire, so a solution to control the internal pressure of the battery cell is required.

SUMMARY

[0005] One aspect of embodiments of the present disclosure includes a battery cell with increased stability by adjusting the internal pressure.

[0006] Another aspect of embodiments of the present disclosure includes a battery cell comprising a vent notch portion having a uniform depth.

[0007] Another aspect of embodiments of the present disclosure includes a method of manufacturing a battery cell case with improved production efficiency by manufacturing a vent notch portion integrally with the case.

[0008] Various embodiments of the present disclosure can be widely applied in the green technology fields such as electric vehicles, battery charging stations, energy storage systems (ESSs), and other technologies using batteries such as photovoltaics and wind power.

[0009] In addition, various embodiments of the present disclosure can also be used for eco-friendly mobility, including electric and hybrid vehicles, to reduce air pollution and greenhouse gas emissions to prevent or reduce climate change.

[0010] A battery cell according to embodiments of the present disclosure includes a case body accommodating an electrode assembly, a case cover covering at least one face of the case body which is open, and a vent notch portion formed by being recessed at at least one of one surface of the case body or one surface of the case cover. The vent notch portion includes a straight portion having a straight pattern, and a curved portion connected to the straight portion and having a curved pattern. A width of the straight portion is different from a width of the curved portion.

[0011] According to an embodiment, the width of the straight portion may be smaller than the width of the curved portion at the one surface of the case body or the one surface of the case cover at which the vent notch portion is formed.

[0012] According to an embodiment, a depth of the straight portion may be the same as a depth of the curved portion.

[0013] According to an embodiment, the vent notch portion may further include a solidification region forming a surface of the straight portion and a surface of the curved portion.

[0014] According to an embodiment, a color of the solidification region may be different from a color of at least one of one surface of the case body other than the surface of the straight portion and the surface of the curved portion, or the one surface of the case cover.

[0015] According to an embodiment, the vent notch portion may further include a connecting portion arranged between the straight portion and the curved portion and connecting the straight

portion and the curved portion. A width of the connecting portion may be different from the width of the straight portion and the width of the curved portion.
[0016] According to an embodiment, ten-point average roughness of a surface of the vent notch portion may be higher than ten-point average roughness of at least one of the one surface of the case body or the one surface of the case cover.
[0017] According to an embodiment, the vent notch portion may further include a bottom portion forming a bottom surface of at least one of the straight portion or the curved portion, and valleys recessed deeper than the bottom portion at opposite sides of the bottom portion.
[0018] According to an embodiment, the vent notch portion may further include a groove recessed inwardly from an outer surface of at least one of the one surface of the case body or the one surface of the case cover.
[0019] According to an embodiment, a surface of the case body or a surface of the case cover which defines the groove may include a plurality of grains. The plurality of grains may be formed in an equiaxed structure.
[0020] According to an embodiment, the vent notch portion may be arranged at an outer surface of one surface of the case body which faces the case cover.
[0021] A method of manufacturing a battery cell case including a case body which accommodates an electrode assembly and a case cover which covers at least one face of the case body which is open according to embodiments of the present disclosure includes forming a vent notch portion including a straight portion and a curved portion at one surface of the battery cell case. The straight portion has a straight pattern and the curved portion is connected to the straight portion and has a curved pattern. A width of the straight portion is smaller than a width of the curved portion. A depth of the straight portion is the same as a depth of the curved portion.
[0022] According to an embodiment, forming the vent notch portion may include etching, using a laser, the one surface of the battery cell case in a first direction in which the straight pattern extends and a second direction which is perpendicular to the first direction.
[0023] According to an embodiment, in the first direction, a period in which the laser intersects the straight pattern may be greater than a period in which the laser intersects the curved pattern.
[0024] According to an embodiment, the method may further include, before forming the vent notch portion, heat treating a portion of the one surface of the battery cell case at which the vent notch portion is formed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0025] FIG. **1** is a diagram illustrating a battery cell according to an embodiment of the present disclosure;
[0026] FIG. **2** is a diagram illustrating a vent notch portion of FIG. **1**;
[0027] FIG. **3** includes cross-sectional views taken along lines A-A and B-B of FIG. **2**;
[0028] FIG. **4** is a diagram illustrating a vent notch portion and a case according to an embodiment of the present disclosure;
[0029] FIG. **5** is a diagram illustrating a vent notch portion and a case according to another embodiment of the present disclosure;
[0030] FIG. **6** is a diagram illustrating a vent notch portion according to another embodiment of the present disclosure;
[0031] FIG. **7** is a diagram illustrating a vent notch portion according to another embodiment of the present disclosure;
[0032] FIG. **8** is a diagram illustrating a method of manufacturing a vent notch portion according to an embodiment of the present disclosure; and

[0033] FIG. **9** is a diagram illustrating a battery cell according to another embodiment of the present disclosure.

DETAILED DESCRIPTION

[0034] Specific structural and functional features of the present disclosure are disclosed in the context of the following embodiments of the disclosure. However, the present disclosure may be configured, arranged, or carried out differently than disclosed herein. Thus, the present disclosure is not limited to any particular embodiment nor to any specific details. Also, throughout the specification, reference to "an embodiment," "another embodiment" or the like is not necessarily to only one embodiment, and different references to any such phrase are not necessarily to the same embodiment. Moreover, the use of an indefinite article (i.e., "a" or "an") means one or more, unless it is clear that only one is intended. Similarly, terms "comprising," "including," "having" and the like, when used herein, do not preclude the existence or addition of one or more other elements in addition to the stated element(s).

[0035] It should be understood that the drawings are simplified schematic illustrations of the described devices and may not include well known details for avoiding obscuring the features of the invention.

[0036] It should also be noted that features present in one embodiment may be used with one or more features of another embodiment without departing from the scope of the invention.

[0037] It is further noted, that in the various drawings, like reference numbers designate like elements.

[0038] FIG. **1** is a diagram illustrating a battery cell **100** according to an embodiment of the present disclosure.

[0039] Referring to FIG. **1**, the battery cell **100** according to an embodiment of the present disclosure may include a case including a case body **110** and a case cover **130**.

[0040] The case body **110** and the case cover **130** may form an internal space in which an electrode assembly is accommodated. The case body **110** may include a conductive metal, such as aluminum, an aluminum alloy, or nickel-plated steel. The case body **110** may have a hexahedral shape including an opening into which the electrode assembly may be inserted. However, embodiments are not limited to the embodiment shown in FIG. **1**. The shape of the case body **110** can be variously modified according to embodiments.

[0041] The case cover **130** may be coupled to the case body **110**. The case cover **130** may be located over the opening in the case body **110**. The case cover **130** may close the opening of the case body **110**. The case cover **130** may cover the opening of the case body **110** to form the internal space in which the electrode assembly is accommodated. The case cover **130** may include an injection hole **133**.

[0042] The electrode assembly may be located in the internal space of the case body **110**. The electrode assembly may include a cathode, a separator, and an anode. The electrode assembly can be in various shapes as needed, such as a stacked or wound shape.

[0043] The cathode may include a cathode current collector. The anode may include an anode current collector. The current collector may include any known conductive material, provided that it does not cause a chemical reaction within a lithium secondary battery. For example, the current collector may include one of stainless steel, nickel (Ni), aluminum (Al), titanium (Ti), copper (Cu), or an alloy thereof, and may be provided in various forms, such as film, sheet, or foil.

[0044] The cathode and the anode can include an active material. The cathode may include a cathode active material and the anode may include an anode active material. The cathode active material may be a material into which lithium ions can be inserted and removed, and the anode active material may be a material into which lithium ions can be absorbed and removed. For example, the cathode active material may be a lithium metal oxide, and the anode active material may be one of carbon-based materials such as crystalline carbon, amorphous carbon, carbon composites, carbon fibers, lithium alloys, silicon (Si), or tin (Sn).

[0045] Each of the cathode and the anode may further include a binder and a conductive material for improved mechanical stability and electrical conductivity.
[0046] The separator may be configured to prevent or mitigate an electrical short between the cathode and the anode and allow the flow of ions to occur. The type of separator is not particularly limited, but may include a porous polymeric film. For example, the separator may include a porous polymeric film or a porous nonwoven fabric. In another example, the separator may include polyethylene, polypropylene, or a composite film of polyethylene and polypropylene.
[0047] An electrolyte may include an electrolyte solution. The electrolyte solution may be a non-aqueous electrolyte solution. The electrolyte solution may include a lithium salt and an organic solvent. The electrolyte solution may further include an additive. The additive may form a film on the cathode or the anode through a chemical reaction in the battery. For example, a cathode interface film may be formed on the cathode and an anode interface film may be formed on the anode.
[0048] After the case body **110** and the case cover **130** are coupled, the electrolyte solution may be injected into the case body **110** through the injection hole **133**. The injection hole **133** may be sealed by a sealing stopper after the electrolyte solution is injected.
[0049] A vent notch portion **150** may be arranged on one surface of the case including the case body **110** and the case cover **130** to allow the adjustment of the internal pressure of the battery cell **100**. The vent notch portion **150** may be formed by being recessed at at least one of one surface of the case body **110** or one surface of the case cover **130**.
[0050] For example, the vent notch portion **150** may be arranged on one surface of the case body **110**. In another example, the vent notch portion **150** may be arranged on one surface of the case cover **130**. When the internal pressure of the battery cell **100** reaches the predetermined pressure, the vent notch portion **150** may be cut to open the vent notch portion **150**. The vent notch portion **150** may include a groove recessed from an outer surface of the case toward the inside of the case. For example, the vent notch portion **150** may include a groove recessed inwardly from an outer surface of at least one of one surface of the case body **110** or one surface of the case cover **130**.
[0051] In FIG. **1**, the vent notch portion **150** may be arranged at an outer surface of one surface of the case body **110** that faces the case cover **130**, but embodiments are not limited thereto. For example, the vent notch portion **150** may be arranged at the case cover **130**. Hereinafter, the vent notch portion **150** is described as being arranged at the case body **110**. Accordingly, while the following description is provided relative to the case body **110**, when the vent notch portion **150** is arranged at the case cover **130**, the description of the case body **110** with respect to the vent notch portion **150** may apply substantially the same to the case cover **130**.
[0052] External terminals **131** may be arranged on the case cover **130**. The external terminals **131** may be spaced apart from each other in a first direction X on one surface of the case cover **130**. The external terminals **131** may be exposed at the top of the case cover **130**. The external terminals **131** may connect the electrode assembly accommodated in the case body **110** to an external power source.
[0053] FIG. **2** is a diagram illustrating the vent notch portion **150** of FIG. **1**.
[0054] Referring to FIG. **2**, the vent notch portion **150** may include a straight portion **151** and a curved portion **152**. The straight portion **151** may have a straight pattern and the curved portion **152** may have a curved pattern.
[0055] The straight portion **151** may be arranged between a plurality of curved portions **152**, connecting the plurality of curved portions **152** to each other. The straight portion **151** may extend between the plurality of curved portions **152**. For example, the straight portion **151** may extend in the first direction X from one surface of the case body **110**.
[0056] The straight portion **151** may have a first width W**151**. The first width W**151** may refer to a width in a direction perpendicular to the direction in which the straight portion **151** extends, i.e., the first width W**151** of the straight portion **151** may refer to the width of the straight portion **151** in

a second direction Y.

[0057] The curved portion **152** may be connected to the straight portion **151**. The curved portion **152** may be connected to each of opposite ends of the straight portion **151** in the direction in which the straight portion **151** extends. In a plan view, the curved portion **152** may have a curved shape, rather than extending in a straight line in a particular direction.

[0058] The curved portion **152** may have a shape that curves away from one end of the straight portion **151** at opposite sides. For example, one curved portion **152** connected to one end of the straight portion **151** may have a semicircular ring shape that is symmetrical with respect to the straight portion **151**, i.e., the one end of the straight portion **151** may be connected to the center of the semicircular ring shape of the curved portion **152**. The plurality of curved portions **152** connected to opposite ends of the straight portion **151** may have a semicircular ring shape that is symmetrical to each other in the direction perpendicular to the direction in which the straight portion **151** extends.

[0059] The curved portion **152** may have a second width W**152**. The second width W**152** may refer to a width in a direction perpendicular to a direction of extension of a tangent line at one point on a boundary of the curved portion **152**.

[0060] The first width W**151** of the straight portion **151** and the second width W**152** of the curved portion **152** may be different. For example, on one surface of the case body **110** at which the vent notch portion **150** is formed, the first width W**151** of the straight portion **151** may be smaller than the second width W**152** of the curved portion **152**.

[0061] FIG. **3** includes cross-sectional views taken along lines A-A and B-B of FIG. **2**.

[0062] Referring to FIG. **3**, the straight portion **151** and the curved portion **152** may include grooves that recessed from one surface **110**S of a case body. The one surface **110**S of the case body may be the outer surface of the case body **110**.

[0063] The first width W**151** of the straight portion **151** may be a width of the straight portion **151** on the one surface **110**S of the case body. The first width W**151** may be the maximum width of the straight portion **151**, but embodiments are not limited thereto.

[0064] The second width W**152** of the curved portion **152** may be a width of the curved portion **152** at the one surface **110**S of the case body. The second width W**152** may be the maximum width of the curved portion **152**, but embodiments are not limited thereto.

[0065] The first width W**151** of the straight portion may be different from the second width W**152** of the curved portion. On the one surface **110**S of the case body, the first width W**151** of the straight portion **151** may be smaller than the second width W**152** of the curved portion **152**, i.e., a width of a mouth of the groove included in the straight portion **151** may be smaller than a width of a mouth of the groove included in the curved portion **152**.

[0066] Each of the straight portion **151** and the curved portion **152** may have a shape that gradually decreases in width along the depth direction from the one surface **110**S of the case body. However, embodiments are not limited thereto. For example, the width of each of the straight portion **151** and the curved portion **152** at a midpoint in the depth direction of the straight portion **151** and the curved portion **152** may be greater than the width of each of the straight portion **151** and the curved portion **152** at the one surface **110**S of the case body.

[0067] The depth direction of the straight portion **151** and the curved portion **152** may refer to a direction perpendicular to the one surface **110**S of the case body at which the straight portion **151** and the curved portion **152** are formed. For example, when the one surface **110**S of the case body at which the straight portion **151** and the curved portion **152** are formed is a plane defined by the first direction X in FIG. **2** and the second direction Y in FIG. **2**, the depth direction of the straight portion **151** and the curved portion **152** may be a third direction Z in FIG. **1**.

[0068] The straight portion **151** may have a first depth D**151**. For example, the first depth D**151** of the straight portion **151** may refer to a depth from the one surface **110**S of the case body at which the straight portion **151** is formed to the maximum depth point of the straight portion **151**. The curved portion **152** may have a second depth D**152**. For example, the second depth D**152** of the curved portion **152** may refer to a depth from the one surface **110**S of the case body at which the curved portion **152** is formed to the maximum depth point of the curved portion **152**.

[0069] The first depth D**151** of the straight portion **151** may be the same as the second depth D**152** of the curved portion **152**, i.e., the maximum depth point of the straight portion **151** and the maximum depth point of the curved portion **152** may be arranged at substantially the same depth from the one surface **110**S of the case body.

[0070] FIG. **4** is a diagram illustrating the vent notch portion **150** and the case according to an embodiment of the present disclosure.

[0071] For reference, FIG. **4** is a schematic illustration of a cross-section of the case taken by Electron Back Scatter Diffraction (EBSD). The EBSD is a method of analyzing the orientation of a material by detecting the electrons that are reflected when accelerated electrons are injected into a sample using a scanning electron microscope equipped with a backscatter electron diffraction pattern analyzer, which allows analysis of the crystal structure of the case.

[0072] Referring to FIG. **4**, the case body **110** may include a plurality of grains **110**G. The plurality of grains **110**G may be formed in an equiaxed structure. For example, a surface of the case body **110** that defines the recessed groove of the vent notch portion **150** may include the plurality of grains **110**G formed in an equiaxed structure, i.e., the plurality of grains **110**G may be non-directional and have similar lengths about a plurality of axes.

[0073] The above-described feature may be attributed to the heating of the case body **110** during the process of forming the recessed groove of the vent notch portion **150**, causing the plurality of grains **110**G to grow side-by-side with respect to a plurality of axial directions. Before forming the recessed groove of the vent notch portion **150**, a portion of the case body **110** at which the vent notch portion **150** is formed may be heat treated and then laser etched to form the recessed groove of the vent notch portion **150**. Thus, the case body **110** may be heated during the heat treatment process so that the plurality of grains **110**G may be formed in the equiaxed structure.

[0074] The case body **110** may include a first portion **110***a* at which the vent notch portion **150** is formed and a second portion **110***b* at which the vent notch portion **150** is not formed. The first portion **110***a* may overlap the vent notch portion **150**. The second portion **110***b* may not overlap the vent notch portion **150**. The second portion **110***b* may include an area other than the first portion **110***a*.

[0075] The size of the plurality of grains **110**G in the first portion **110***a* and the size of the plurality of grains **110**G in the second portion **110***b* may be uniform. For example, the standard deviation of the size of the plurality of grains **110**G in the first portion **110***a* and the size of the plurality of grains **110**G in the second portion **110***b* may be **20** or less. Because the vent notch portion **150** is etched using a laser without pressurizing the case body **110**, the size of the plurality of grains **110**G may be uniform across a portion of the case body **110** where the vent notch portion **150** is formed and a portion where the vent notch portion **150** is not formed.

[0076] The vent notch portion **150** may include a solidification region **155**. The solidification region **155** may form a surface of the vent notch portion **150**. For example, the solidification region **155** may form a surface of the straight portion **151** in FIG. **3** and the curved portion **152** in FIG. **3**. The solidification region **155** may be formed as the case body **110** is etched using a laser to form the vent notch portion **150**. The solidification region **155** may be formed as the case body **110** is heated and melted by the laser and then re-solidified.

[0077] The color of the solidification region **155** may be different from the color of a surface of the case body **110** on which the vent notch portion **150** is not formed. For example, the color of the solidification region **155** may be different from the color of a surface of the case body **110** other than the surface of the vent notch portion **150**. The color of the solidification region **155** may be different from the color of one surface of the case body **110** other than the surface of the straight portion **151** in FIG. **3** and the surface of the curved portion **152** in FIG. **3**. For example, the color of the solidification region **155** may be darker than the color of the surface of the case body **110** at which the vent notch portion **150** is not formed, which can be attributed to the discoloration of the solidification region **155** as the case body **110** is heated and melted by the laser and then re-solidified.

[0078] The ten-point average roughness Rz of the surface of the vent notch portion **150** may be higher than the ten-point average roughness Rz of the surface of the case body **110** other than the vent notch portion **150**. The surface of the case body **110** other than the vent notch portion **150** may refer to an outer surface. The ten-point average roughness Rz of the surface of the vent notch portion **150** may be higher than the ten-point average roughness of one surface of the case body **110**.

[0079] The ten-point average roughness is a numerical value of the roughness of the surface. Known methods of measuring roughness can be used to measure the roughness of the surface of the vent notch portion **150** and the surface of the case body **110**. As a result, the measured ten-point average roughness of the surface of the vent notch portion **150** may be higher than the ten-point average roughness of the surface of the case body **110** where the vent notch portion **150** is not formed. In other words, the surface of the vent notch portion **150** may be rougher than the surface of the case body **110** where the vent notch portion **150** is not formed.

[0080] This can be attributed to the fact that the vent notch portion **150** is formed using a laser. When the vent notch portion **150** is formed by a pressurization method but not a laser, the surface of the vent notch portion **150** pressurized may be flat. On the other hand, when the vent notch portion **150** is formed by a laser, the surface of the vent notch portion **150** may be rough during the etching process. For example, the ten-point average roughness of the surface of the vent notch portion **150** may exceed **20**.

[0081] The vent notch portion **150** may include a bottom portion **150***a*. The bottom portion **150***a* may form a bottom surface of the vent notch portion **150**. For example, the bottom portion **150***a* may form a bottom surface of at least one of the straight portion **151** in FIG. **3** or the curved portion **152** in FIG. **3**. The bottom portion **150***a* may include the bottom surface of the straight portion **151** of FIG. **3**. The bottom portion **150***a* may include the bottom surface of the curved portion **152** in FIG. **3**.

[0082] FIG. **5** is a diagram illustrating the vent notch portion **150** and the case according to another embodiment of the present disclosure. By way of example, differences from those described with reference to FIG. **4** will be mainly described.

[0083] Referring to FIG. **5**, the vent notch portion **150** may include the bottom portion **150***a* and valleys **150***b*. The valleys **150***b* may be recessed deeper than the bottom portion **150***a* at opposite sides of the bottom portion **150***a*. The valleys **150***b* may be arranged at opposite sides of the bottom portion **150***a* in the width direction of the vent notch portion **150**, i.e., the depth from one surface of the case body **110** to the valley **150***b* may be greater than the depth from one surface of the case body **110** to the bottom portion **150***a*, which may be attributed to the fact that during the process of forming the vent notch portion **150** with a laser, heat is concentrated at the corners of the groove of the vent notch portion **150**.

[0084] FIG. **6** is a diagram illustrating the vent notch portion **150** according to another embodiment of the present disclosure. By way of example, differences from those described with reference to FIG. **2** will be mainly described.

[0085] Referring to FIG. **6**, the straight portion **151** may be connected to an end that is not in the center of the semicircular ring shape of the curved portion **152**. The curved portion **152** may be bent from opposite ends of the straight portion **151** to one side. One curved portion **152** connected to one end of the straight portion **151** may not be symmetrical with respect to the straight portion **151**, i.e., the curved portion **152** may be arranged at one side only and not on the other side relative to the direction in which the straight portion **151** extends.

[0086] FIG. **7** is a diagram illustrating the vent notch portion **150** according to another embodiment of the present disclosure. By way of example, differences from those shown in FIG. **2** will be mainly described.

[0087] Referring to FIG. **7**, the vent notch portion **150** may include a connecting portion **153** connecting the straight portion **151** and the curved portion **152**. The connecting portion **153** may be arranged between the straight portion **151** and the curved portion **152**.

[0088] The connecting portion **153** may have a third width W**153**. The third width W**153** of the connecting portion **153** may be different from the first width W**151** of the straight portion **151** and the second width W**152** of the curved portion **152**. For example, the third width W**153** of the connecting portion **153** may be smaller than each of the first width W**151** of the straight portion **151** and the second width W**152** of the curved portion **152**. In another example, the third width W**153** of the connecting portion **153** may be greater than the first width W**151** of the straight portion **151** and smaller than the second width W**152** of the curved portion **152**. The depth of the connecting portion **153** may be the same as the depth of the straight portion **151** and the depth of the curved portion **152**.

[0089] In FIG. **7**, it is illustrated that the connecting portion **153** has a straight line pattern, but embodiments are not limited thereto. For example, the connecting portion **153** may have a curved pattern.

[0090] FIG. **8** is a diagram illustrating a method of manufacturing the vent notch portion **150** according to an embodiment of the present disclosure. By way of example, differences from those described with reference to FIG. **2** will be mainly described.

[0091] Referring to FIG. **8**, a method of manufacturing a battery cell case may include forming, at one surface of the battery cell case, the vent notch portion **150** including the straight portion **151** having a straight pattern and the curved portion **152** connected to the straight portion **151** and having a curved pattern. Further, the method of manufacturing the battery cell case may include, before forming the vent notch portion **150**, heat treating a portion of one surface of the battery cell case at which the vent notch portion **150** is formed.

[0092] The vent notch portion **150** may be formed by laser etching one surface of the case body **110** of FIG. **1** or one surface of the case cover **130** of FIG. **1** along a laser pattern LP to form a groove.

[0093] The laser pattern LP may extend in the direction in which the straight portion **151** extends and in a direction perpendicular thereto. For example, the laser pattern LP may include a plurality of straight patterns extending in the first direction X and the second direction Y. In other words, forming the vent notch portion may include etching one surface of the battery cell case with a laser in the first direction X and the second direction Y.

[0094] The laser forming the vent notch portion **150** may etch, by heating and melting, one surface of the case body **110** of FIG. **1** or one surface of the case cover **130** of FIG. **1** at which the vent notch portion **150** is formed, along the laser pattern LP.

[0095] In the first direction X in which the straight portion **151** extends, a period in which the laser intersects the straight pattern of the straight portion **151** along the laser pattern LP may be greater than a period in which the laser intersects the curved pattern of the curved portion **152**. Therefore, in the direction in which the straight portion **151** extends, the curved pattern of the curved portion **152** that the laser intersects more than the straight pattern of the straight portion **151** may be etched relatively more than the straight pattern of the straight portion **151**. Therefore, by making the width of the curved portion **152** larger than the width of the straight portion **151**, the depth of the curved portion **152** and the depth of the straight portion **151** may be uniformly etched.

[0096] FIG. **9** is a diagram illustrating a battery cell according to another embodiment of the present disclosure. By way of example, differences from those described with reference to FIG. **1** will be mainly described.

[0097] Referring to FIG. **9**, the vent notch portion **150** may be arranged at one surface of the case body **110** adjacent to the case cover **130**. The plurality of external terminals **131** may be arranged

on the case covers **130**, each closing an opening face of the case body **110** that is open in the first direction X and facing each other.
[0098] The present disclosure is not limited to the embodiments described above, as the present disclosure may be implemented and modified in various forms. Accordingly, when modifications of the embodiments include components of the present disclosure, such modifications should be considered to fall within the scope of the present disclosure.
[0099] According to some embodiments of the present disclosure, the internal pressure of a battery cell may be adjusted, thereby improving the stability.
[0100] According to some embodiments of the present disclosure, the production efficiency of the battery cell may be improved.
[0101] According to some embodiments of the present disclosure, the production efficiency may be improved by manufacturing a vent notch portion integrally with a case.

Claims

1. A battery cell, comprising: a case body accommodating an electrode assembly; a case cover covering at least one face of the case body which is open; and a vent notch portion formed by being recessed at at least one of one surface of the case body or one surface of the case cover, wherein the vent notch portion includes: a straight portion having a straight pattern; and a curved portion connected to the straight portion and having a curved pattern, and wherein a width of the straight portion is different from a width of the curved portion.
2. The battery cell according to claim 1, wherein the width of the straight portion is smaller than the width of the curved portion at the one surface of the case body or the one surface of the case cover at which the vent notch portion is formed.
3. The battery cell according to claim 1, wherein a depth of the straight portion is the same as a depth of the curved portion.
4. The battery cell according to claim 1, wherein the vent notch portion further includes a solidification region forming a surface of the straight portion and a surface of the curved portion.
5. The battery cell according to claim 4, wherein a color of the solidification region is different from a color of at least one of one surface of the case body other than the surface of the straight portion and the surface of the curved portion, or the one surface of the case cover.
6. The battery cell according to claim 1, wherein the vent notch portion further includes a connecting portion arranged between the straight portion and the curved portion and connecting the straight portion and the curved portion, and wherein a width of the connecting portion is different from the width of the straight portion and the width of the curved portion.
7. The battery cell according to claim 1, wherein ten-point average roughness of a surface of the vent notch portion is higher than ten-point average roughness of at least one of the one surface of the case body or the one surface of the case cover.
8. The battery cell according to claim 1, wherein the vent notch portion further includes: a bottom portion forming a bottom surface of at least one of the straight portion or the curved portion; and valleys recessed deeper than the bottom portion at opposite sides of the bottom portion.
9. The battery cell according to claim 1, wherein the vent notch portion further includes a groove recessed inwardly from an outer surface of at least one of the one surface of the case body or the one surface of the case cover.
10. The battery cell according to claim 9, wherein a surface of the case body or a surface of the case cover which defines the groove includes a plurality of grains, and wherein the plurality of grains are formed in an equiaxed structure.
11. The battery cell according to claim 1, wherein the vent notch portion is arranged at an outer surface of one surface of the case body which faces the case cover.
12. A method of manufacturing a battery cell case including a case body which accommodates an

electrode assembly and a case cover which covers at least one surface of the case body which is open, the method comprising: forming a vent notch portion including a straight portion and a curved portion at one surface of the battery cell case, wherein the straight portion has a straight pattern and the curved portion is connected to the straight portion and has a curved pattern, wherein a width of the straight portion is smaller than a width of the curved portion, and wherein a depth of the straight portion is the same as a depth of the curved portion.

13. The method according to claim 12, wherein forming the vent notch portion includes: etching, using a laser, the one surface of the battery cell case in a first direction in which the straight pattern extends and a second direction which is perpendicular to the first direction.

14. The method according to claim 13, wherein, in the first direction, a period in which the laser intersects the straight pattern is greater than a period in which the laser intersects the curved pattern.

15. The method according to claim 12, further comprising, before forming the vent notch portion, heat treating a portion of the one surface of the battery cell case at which the vent notch portion is formed.
